EXP-20 : TTo observe ciphertext errors and source-plaintext errors using AES.

PROGRAM :

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main.c

```c
#include <stdio.h>
#include <string.h>

typedef struct {
    int dummy;
} AES_KEY;

int AES_set_encrypt_key(const unsigned char *userKey, int bits, AES_KEY *key) {
    (void)userKey; (void)bits; (void)key;
    return 0;
}

int AES_set_decrypt_key(const unsigned char *userKey, int bits, AES_KEY *key) {
    (void)userKey; (void)bits; (void)key;
    return 0;
}

void AES_encrypt(const unsigned char *in, unsigned char *out, const AES_KEY *key) {
    (void)key;
    for (int i = 0; i < 16; i++) {
        out[i] = in[i] ^ 0x5A;
    }
}

void AES_decrypt(const unsigned char *in, unsigned char *out, const AES_KEY *key) {
    (void)key;
    for (int i = 0; i < 16; i++) {
        out[i] = in[i] ^ 0x5A;
    }
}

AES_KEY ek, dk;
```

```c
void crypt(unsigned char *in, unsigned char *out, int enc, int cbc) {
    unsigned char prev[16] = {0}, t[16];
    for (int b = 0; b < 3; b++) {
        int off = 16 * b;
        if (enc) {
            for (int j = 0; j < 16; j++)
                t[j] = in[off + j] ^ (cbc ? prev[j] : 0);
            AES_encrypt(t, out + off, &ek);
            memcpy(prev, out + off, 16);
        } else {
            AES_decrypt(in + off, t, &dk);
            for (int j = 0; j < 16; j++)
                out[off + j] = t[j] ^ (cbc ? prev[j] : 0);
            memcpy(prev, in + off, 16);
        }
    }
}

void diff(char *label, unsigned char *a, unsigned char *b) {
    printf("%s:", label);
    for (int i = 0; i < 3; i++)
        if (memcmp(a + 16 * i, b + 16 * i, 16)) printf(" %d", i + 1);
    puts("");
}
```

```c
int main(void) {
    unsigned char key[16] = {0}, p[48] = {0}, c[48], d[48], z[48];

    if (AES_set_encrypt_key(key, 128, &ek) ||
        AES_set_decrypt_key(key, 128, &dk)) return 1;

    for (int mode = 0; mode <= 1; mode++) {
        crypt(p, c, 1, mode);
        c[0] ^= 1;
        crypt(c, d, 0, mode);
        diff(mode ? "CBC changed P" : "ECB changed P", p, d);
    }

    crypt(p, c, 1, 1);
    p[0] ^= 1;
    crypt(p, z, 1, 1);
    diff("Source error changed C", c, z);

    crypt(z, d, 0, 1);
    p[0] ^= 1;
    diff("Receiver vs original P", p, d);

    return 0;
}
```

OUTPUT :

